Anirudh Chandan

Backend Software Engineer · Python (FastAPI) · PostgreSQL · Document pipelines & applied AI

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SUMMARY

Backend engineer (2.5+ years incl. freelance) shipping **production Python (FastAPI, Flask) with PostgreSQL, Celery and Redis**, plus Node.js and Spring Boot. Builds **document pipelines with AI kept on a leash**: classification that never guesses, rule-defined cross-document validation, opt-in AI lanes behind human confirmation. Owns features end to end, data model to rollout. Built **PyDB, a transactional storage engine from scratch** (B-Trees, write-ahead log, crash recovery). Daily Claude Code user. 450+ LeetCode.

EXPERIENCE

Independent Software Contractor *Aug 2026 – Present*
Backend / Full-Stack Engineer (freelance) *Remote · India / Australia*

- Built a **multi-tenant document workflow platform** (Python/FastAPI, PostgreSQL row-level security, Celery, Alembic) that combines **unstructured documents with structured business data**: layered classification, YAML cross-document validation rules (e.g. bill-of-lading weights vs manifest ±0.5%), maker-checker approval with human override, immutable versions, **two opt-in AI lanes (off by default) and a human-gated learning normaliser (suggest → confirm → revoke)**, 400+ pytest tests; generalised into pluggable industry packs (shipping, **procure-to-pay with 3-way match**).
- Delivered a **live AI-CCTV alert dashboard** (Node.js) for a gated villa community: ingests events from an on-prem AI box via HTTP callbacks, patrol-compliance timers, EN/HI voice alerts, role-based remote access, Excel/PDF reports — **in production since Sep 2026**.
- Electricity-theft screening POC (state power distributor): **7 versioned YAML rules + an Isolation Forest anomaly layer** over ~5.5M meter readings scored in ~10 s; field verdicts feed a shadow-train loop.

Docplix *Nov 2025 – Present*
Software Engineer *Remote*

- Built and led 30+ Node.js/Sequelize REST APIs for an EHR platform serving **2,300+ clinics**, as one of **4 engineers**; ran a **zero-downtime v1→v2 migration with 100% data integrity**.
- Designed **optimistic concurrency control and distributed Redis locks** for the central sync service, eliminating race conditions behind concurrent-write data mismatches.

Genpact *Feb 2024 – Oct 2025*
Software Engineer (promoted from Intern) *Bengaluru, India*

- Engineered **idempotent, exactly-once processing** for a Kafka tax-ingestion pipeline using database constraints, guaranteeing correctness across partitions during network failures.
- Built resilient Python (Flask) microservices with automated retries, timeouts and dead-letter queues, serving 24 modules with **zero data loss**; production Python that real customers depended on since 2024.
- Integrated structured logging and **distributed tracing** across serverless GCP Cloud Functions, cutting MTTR for critical production bugs by **40%**.

PROJECTS

PyDB — Database Storage Engine Built From Scratch. A disk-based transactional storage engine in pure Python: its own **pager and 8 KB page cache**, raw binary serialization, **clustered and secondary B-Tree indexes**, and a **write-ahead log** with redo recovery and atomic checkpoints. Recovery is tested by **SIGKILLing a writer mid-workload**; CI on every push. (live demo · code)

ProjAuto — Multi-tenant SaaS ERP for project & procurement workflows (20+ live tenants, 120+ services). Top contributor (711 of 1,100 commits): migrated a legacy Java monolith to **Spring Boot + React**, eliminated N+1 queries across 53 endpoints (**15.6s → 2.4s**), built a distributed token-bucket rate limiter, and closed account-takeover and cross-tenant IDOR security holes.

SKILLS

Backend: Python (FastAPI, Flask, SQLAlchemy, Alembic, Celery, pytest), Node.js/TypeScript (Express, Sequelize), Java (Spring Boot), REST API design (versioning, pagination, validation), modular monolith vs microservices trade-offs

Data & Messaging: PostgreSQL (row-level security, query plans, indexing, migrations under load), MySQL, Redis (caching, Pub/Sub, locks), Apache Kafka, BullMQ, database internals (B-Trees, WAL, ACID)

Cloud & DevOps: AWS, GCP (Cloud Functions / serverless), Docker, Linux, Git, CI/CD (GitHub Actions)

Practices: end-to-end feature ownership, idempotency & retries, TDD-style behaviour tests, root-cause debugging with regression tests, distributed tracing, structured logging, AI-assisted development (Claude Code, Codex)

Foundations: Data Structures & Algorithms (450+ LeetCode), OOP, SQL

EDUCATION & ACHIEVEMENTS

Graphic Era University, Dehradun *2020 – 2024*

B.Tech, Computer Science & Engineering — CGPA 9.11

Technical blog on systems internals (anirudhchandan.com/blog).